

PASS 2 CONSTRUCTIVE APPENDIX REVISION PACKAGE

for "Compressed Odd Dynamics and a Structural Proof of the Collatz Conjecture"

Status note

This package hardens the manuscript structure and provides Appendix A scaffolding, CSV schemas, and a certification manifest template. It does not certify the theorem or claim that the elimination tables are complete or verified.

Appendix A replacement text

This appendix is logically load-bearing only after the template lists and row-wise elimination reports are fully generated and independently checked.

We work with the constant package $X_0 = 7$, $Y_0 = 12$, $\epsilon_0 > 0$, $M = 2^{14}$.

For T_R , search lengths satisfy $2 \leq m \leq m_R$. For C_{nc} , search lengths satisfy $1 \leq m \leq m_C$. These are certified bounded-length cutoffs, not editorial guesses.

Enumeration of T_R : enumerate all words with $7 \leq \beta_i \leq Y_0$, compute $B(T)$, $S_r(T)$, $A(T)$, $C(T)$, retain only those with $A(T) < 1$ and $1 - A(T) \leq C(T) < 1$, then apply deep admissibility and residue constraints.

Enumeration of C_{nc} : enumerate all words with $2 \leq \beta_i \leq Y_0$, compute $B(T)$, $B_{ren}(T) = \sum(\beta_i - 1)$, $\Delta_{ren}(T) = B_{ren}(T) - (\log_2 3 - 1)m$, compute $A(T)$, $C(T)$, retain only those satisfying $A(T) < 1$, $C(T) \geq 1 - A(T)$, $|\Delta_{ren}(T)| \leq \epsilon_0$, and local admissibility constraints.

Each T_R row must record the first failed test among divisibility, positivity/oddness, forward valuation compatibility, residue recurrence, and strict contraction.

Each C_{nc} row must record the first failed test among divisibility, positivity/oddness, forward valuation compatibility, residue recurrence, renormalized drift compatibility, and additive compensation compatibility.

Main-text integration map

Theorem 6.3 (elimination of R) should cite Theorem A.6.

Theorem 7.10 (bounded-length theorem) should cite an analytic bound theorem or Proposition A.2.

Proposition 7.11 (finite endgame reduction) should cite Propositions A.2 and A.9.

Theorem 7.5 (elimination of C) should cite Theorem A.11.

Next step

Populate the CSV files with actual finite lists and elimination rows from a settled search definition. Until then, this package remains a constructive certification framework rather than a completed proof appendix.

CSV table schemas

TR_templates.csv

id	m	beta_wor d	B	A	C	deep_ad missible	residue_a dmissible	status
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Cnc_templates.csv

id	m	beta_wo rd	B	B_ren	Delta_r en	A	C	local_ad missible	status
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TR_elimination_report.csv

id	D_T	P_T	divisibl e	n0	positive _odd	valuatio n_ok	residue_ ok	strict_c ontracti on	first_fai lure
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Cnc_elimination_report.csv

id	D_T	P_T	divisibl e	n0	positiv e_odd	valuati on_ok	residue _ok	drift_o k	compe nsation _ok	first_fa ilure
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Certification manifest template

```
{
  "constants": {
    "X0": 7,
    "Y0": 12,
    "epsilon0": "",
    "M": 16384
  },
  "cutoffs": {
    "m_R": "",
    "m_C": ""
  },
  "files": {
    "TR_templates.csv": "sha256:",
    "Cnc_templates.csv": "sha256:",
    "TR_elimination_report.csv": "sha256:",
    "Cnc_elimination_report.csv": "sha256:"
  },
  "code_version": "",
  "status": "draft"
}
```